

Productivity enhancers for neurodivergent adults

- **Task management.** Google Calendar, Microsoft Outlook, Todoist, Trello, Sunsama, Notion, Evernote, and Reclaim.ai
- **Communications.** Grammarly, Hemingway Editor, Otter, Rev, Speechnotes, Dragon NaturallySpeaking, Google Translate, Microsoft Immersive Reader, Speechify, MindMeister, and Glean
- **Miscellaneous tasks.** Goblin Tools, GPT-4, and Claude

recommending on their blogs. Goblin Tools defines itself as ‘a collection of small, simple tools, for when things feel too big or complicated’—and that speaks volumes. It has tools to break down tasks into smaller ones, help with decision-making, turn a brain dump into actionable items, create a recipe, read aloud, or explain something, and more. Then, of course, there are neurodiversity-friendly chatbots like GPT-4 and Claude that help with just about everything.

Need for caution and guardrails

As in all applications of AI, here too the entire ASD community—from doctors and therapists to parents, other caregivers, and adult users themselves, need to be extremely cautious to prevent imprudent usage and over-reliance on technology. Use of AI in AAC and other assistive technologies is still nascent, and has lot of wrinkles that need to be ironed out. In some cases, the apps may be developed without proper expert advice too. So, one should do the due diligence before picking any technologies for neurodivergent individuals.

Here are some challenges that are yet to be overcome in AI-based assistive technologies.

Resource intensive. Integrating diverse data sources like genetic profiles, neuroimaging, and behavioural assessments is complex, requiring advanced algorithms and substantial computational resources. Even though technically we may say that online resources and telehealth

portals can be accessed from anywhere and improve global reach of AI-powered solutions to manage ASD, the high costs of developing and implementing advanced AI tools limit this accessibility. The tech world is still figuring out how to reduce the resource consumption of AI systems.

Inadequacy of technologies.

Sometimes, the quality of technology that one has access to, like the quality of sensors, VR platforms, or headsets, may not be up to the mark. This may disturb the desired outcome. For example, when using AI for diagnosis and progress tracking, therapists and caregivers rely on the accuracy of data collected by sensors, but this may not always be so. Hence, results presented by AI-based systems must be correlated and corroborated frequently with feedback from caregivers.

Similarly, VR and AR systems for simulating real-world scenarios also come with a bunch of problems. Many VR systems are still low-fidelity, and many do not recreate the real-world details accurately enough. Sometimes even colours and sounds can be distorted. Plus, some individuals face discomfort and even motion sickness when wearing VR headsets, which can undermine therapeutic benefits. Designers should focus on the ergonomics of headsets and other wearables for neurodivergent individuals. And caregivers and therapists must select wearables that are suitable for the user. Sadly, many commercial

wearables lack explicit approvals in their countries for ASD-specific applications.

Model failure. AI models can fail in unpredictable real-world situations, or what are called ‘out-of-dataset’ scenarios. They may also develop inappropriate content or training materials if trained on non-representative data. Care should be taken to ensure that content generated by AI apps is ethically, culturally, and linguistically appropriate.

Problems with fit. The inherent diversity in the ASD spectrum makes it difficult to find a perfectly suitable solution for any individual. Even though apps claim a great level of personalisation and on-the-go customisation, even that will be within a framework, which itself might not suit some neurodivergent individuals. Hence, it is important for caregivers to be constantly involved in the lives of autistic children and adults, ensuring they are comfortable with whatever technology tools are being used.

Longitudinal uncertainty.

Although AI-enabled technology appears promising, most of the current research provides only short-term outcomes, and there are very few longitudinal studies that cover the long-term efficacy and impact of AI interventions on the progressive nature of ASD. So, we need to take the promise of AI with a pinch of salt, as always.

Data privacy and security.

Like all AI systems, concerns about privacy, consent, and potential data misuse are also relevant here, as a significant amount of sensitive personal data is constantly collected for tailoring interventions. Developers must implement robust cybersecurity measures. Caregivers must also verify whether sufficient privacy and security measures are